

GIRIDHARAN K

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Summary

Security-focused developer in digital forensics, OSINT tooling, and application security, building Python threat-intelligence and image-forensics systems. Solved 40+ TryHackMe labs, global top 11%. Targeting an entry-level Application Security Analyst or Security Engineer role.

Education

Sri Kirana Arts and Science College

Tamil Nadu, India

M.Sc. Computer Technology, 2026 – 2027

Kongu Arts and Science College, Bharathiar University

Erode, Tamil Nadu

B.Sc. Computer Science and Applications, 2023 – 2026

Technical Skills

Languages: Python, JavaScript (ES6+), TypeScript, Bash, SQL, HTML5/CSS3

Security Tools: Nmap, Burp Suite, Wireshark, Metasploit, OWASP ZAP, SQLmap, Scapy, TShark, Ghidra, BloodHound, Kali Linux

Security Domains: Penetration Testing, OWASP Top 10, MITRE ATT&CK, Web Exploitation (XSS, CSRF, SQLi, SSRF, IDOR), Privilege Escalation, Active Directory Attacks, Incident Response, SOC Triage, Threat Intelligence, IAM/RBAC, Least-Privilege Access, ISO 27001, Critical Infrastructure Protection (ICS/OT), Digital Forensics, OSINT

Frameworks & ML: React.js, Flask, FastAPI, MongoDB, REST APIs, Socket.IO, scikit-learn, XGBoost, SHAP

Cloud & DevOps: AWS (IAM, EC2, S3), Azure (Entra ID, RBAC), Docker, CI/CD, Git/GitHub, Linux CLI

Projects

matazero — Image Forensics & OSINT Toolkit | *Python, JPEG/EXIF Internals, Ollama, SQLite* 2026

- Built a fully offline image forensics CLI with 21 subcommands, running a 7-tier extraction pipeline over JPEG, PNG, TIFF/RAW, WebP, and HEIC evidence
- Attributed images to source hardware after metadata stripping by matching quantization tables, Huffman tables, and chroma subsampling to a 22+ device corpus
- Detected tampering and AI-generated imagery through 3 independent signals: JPEG ghost double-compression, CFA Bayer demosaicing periodicity, and copy-move clone detection
- Enforced chain of custody with SHA-256 hash-chained audit logs, HMAC-signed authorization scopes, and sandboxing

NexShield — AI Threat Intelligence Platform | *Python, FastAPI, MongoDB, scikit-learn* Oct 2025 – Feb 2026

- Built a threat intelligence platform automating Nmap reconnaissance across 28+ sensitive service signatures, mapping findings to MITRE ATT&CK
- Designed a 9-engine analysis pipeline covering port risk scoring, version vulnerability matching (Log4Shell, Heartbleed, SambaCry), and default credential detection
- Trained a Random Forest classifier for threat severity and cached NVD CVE responses in MongoDB, cutting lookups by 80%

DNS Sinkhole — DNS Threat Surveillance Platform | *Python, Scapy, TShark, MongoDB* Dec 2025 – Mar 2026

- Engineered a real-time DNS interception platform with Scapy/TShark dual-engine capture and inline domain quarantine
- Detected Domain Generation Algorithm (DGA) domains through Shannon entropy scoring, plus DNS tunneling patterns and suspicious TLD heuristics
- Extended coverage to 4 transport protocols (DNS-over-TCP, DoT, DoH, DoQ) and added drag-and-drop PCAP forensic replay

Experience

Future Interns

2026

Cyber Security Intern

Remote

- Performed API risk assessments and payload analysis on web application endpoints, documenting injection and authorization weaknesses
- Tracked and classified web threat activity in a simulated environment, mapping findings to OWASP Top 10 categories

Telstra (Forage)

Dec 2025

Cybersecurity Incident Response — Job Simulation

Remote

- Triage simulated incidents via SIEM-style log correlation, tracing malware propagation across network segments
- Reconstructed attack timelines from intrusion vector to lateral movement, documenting remediation

Certifications & Achievements

TryHackMe: 40+ labs in enumeration, web exploitation, privilege escalation, and OSINT; global top 11% of 3M+ users

Certifications: Google Cybersecurity Professional; Cisco Ethical Hacker; Fortinet Network Security Associate; Critical Infrastructure Protection (OPSWAT); AWS Solutions Architecture